

1. (10%) Consider the following code sequence. Is there a loop-carried dependence? Is there any way that we can perform parallel? If Yes, how should we do?  $\gamma es$ ,

```
For (i=1; i<=100; i=i+1){
    A[i]=A[i]+B[i];    //S1
    B[i+1]=C[i]+D[i]; //S2
}
```

• After Rewriting:  
 A[1]=A[1]+B[1];  
 For (i=1; i<=99; i=i+1){  
 B[i+1]=C[i]+D[i];  
 A[i+1]=A[i+1]+B[i+1];  
 }  
 B[101]=C[100]+D[100];

### Latencies and initiation intervals for function units

# of intervening cycles between an instruction that produces a result and an instruction that uses the result.

| Functional unit | Latency (clock cycle) | Initial interval (Repeat interval) |
|-----------------|-----------------------|------------------------------------|
| Integer ALU     | 0                     | 1                                  |
| Data memory     | 1                     | 1                                  |
| FP add          | 3                     | 1                                  |
| FP multiply     | 6                     | 1                                  |
| FP divide       | 24                    | 25                                 |

# of clock cycles that must elapse between issuing two operations of a given type

- As we can observe from the table, EX stage needs not only one cycle

2. (10%) Please explain the latencies and initiation intervals for function units.

3. (10%) Which of the following statements are true for scoreboard?

- (A) Scoreboard allows out-of-order issuing.
- (B) Scoreboard allows out-of-order execution.
- (C) Scoreboard allows the instructions to be completed out of order.
- (D) Scoreboard checks for structural hazard before issuing the instructions.
- (E) Scoreboard checks for WAW hazard when writing back the result.

BCD

4. (10%) Which of the following statements are true for Tomasulo algorithm with hardware based speculation?

- (A) WAR and WAW hazards are eliminated with the help of reordering buffer.
- (B) After an instruction is issued, the operands that are ready would be stored in the reservation station.
- (C) The instructions are allowed to be executed and committed out of order.
- (D) The register file or memory can be updated by an instruction before the instruction commits.
- (E) Dynamic scheduling of different combinations of basic blocks can be performed.

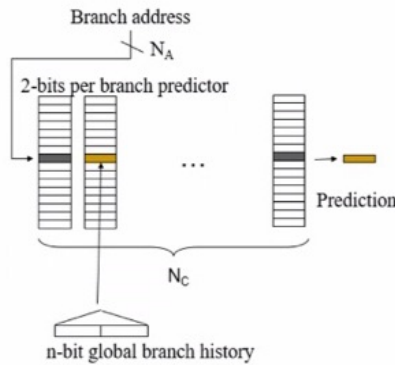
BE

5. (10%) Which of the following statements are true?

- (A) Scoreboard checks for RAW hazard in the read operand stage.
- (B) Scoreboard checks for WAR hazard in the write back stage.
- (C) When a branch with an incorrect prediction commits, the speculation is wrong and the ROB is flushed. The execution is restarted at the correct successor of the branch.
- (D) In Tomasulo algorithm, store buffer holds the components of the effective address until it is computed.
- (E) In Tomasulo algorithm, load buffer holds results of completed loads.

ABCDE

6. (10%) Consider a (3, 2) correlating predictor illustrated in the figure below. Suppose the number of bits in the predictor is 32K. What is the value of  $N_A$ ,  $N_C$ , and  $n$  in the figure, respectively?



$$2^3 \cdot 2^2 = 2^5$$

$$k' = 11$$

$$\therefore N_A = 11$$

$$N_C = 8$$

$$n = 2$$

7. (10%) Consider the following code sequence. Suppose the branch address for branch b3 is ECFF F3AB. Assume the latest behavior for branch b1 is not taken (denoted by 0), and the latest behavior for branch b2 is taken (denoted by 1). If we use a 2 bit correlating branch predictor to predict the behavior of b3, and the content of the branch history table is listed as the figure below, what is the prediction for b3 from the table? Please explain your answer.

Behavior of the last two branches:

```

DSUBUI    R3, R1, #2
BNEZ      R3, L1      ; branch b1
ANDI      R1, R1, #0
L1: SUBUI  R3, R2, #2
BNEZ      R3, L2      ; branch b2
ANDI      R2, R2, #0
L2: SUBU   R3, R1, R2
BEQZ      R3, L3      ; branch b3
    
```

|      | 00 | 01 | 10 | 11 |
|------|----|----|----|----|
| 0000 | 10 | 00 | 11 | 01 |
| 0001 | 11 | 01 | 00 | 11 |
| 0010 | 01 | 01 | 11 | 11 |
| 0011 | 00 | 10 | 11 | 00 |
| 0100 | 00 | 00 | 10 | 01 |
| 0101 | 11 | 01 | 01 | 10 |
| 0110 | 01 | 01 | 11 | 10 |
| 0111 | 10 | 00 | 10 | 11 |
| 1000 | 00 | 11 | 10 | 00 |
| 1001 | 10 | 11 | 01 | 10 |
| 1010 | 01 | 01 | 00 | 11 |
| 1011 | 01 | 11 | 11 | 00 |
| 1100 | 00 | 01 | 10 | 00 |
| 1101 | 10 | 10 | 00 | 11 |
| 1110 | 11 | 01 | 00 | 10 |
| 1111 | 01 | 00 | 11 | 10 |

$b_3$  的 address 是 ECFF F3AB, 看 B = (1011)<sub>2</sub>,  $b_1$  not taken,  $b_2$  taken 所以用 01  
查表後可得 11, 代表  $b_1, b_2$  均 taken, 表示  $R_3 = R_1 - 2 = R_2 - 2 \Rightarrow R_1 = R_2$ ,  
 $\therefore R_3 = R_1 - R_2 = 0$  BEQZ  $R_3, L_3$  會 taken.

8. (10%) Why is dynamic scheduling important?

1. keep compiler simpler 2. compiled code for one machine runs well on another.

9. (10%) Please explain SIMD (Single Instruction stream Multiple Data stream). What are the applications suitable for it?

1. Data level parallelism, 2. Each has its data memory but a single instruction memory and control processor.  
3. multi-media, graphic performance.

(A) SuperScalar allows issuing multiple instructions per cycle and therefore can achieve a CPI that is smaller than 1.

(B) It is impossible for a processor supporting Very Long Instruction Word to achieve a CPI that is smaller than 1.

(C) Strip mining is used to generate code such that each vector operation is done for a size less than or equal to the maximum vector length.

(D) It is possible for programs that contain "if" statements in loops to be vectorized and run in vector mode.

(E) The impact of control hazards is smaller in pipelines with more stages (longer pipelines).

(A) (C) (D)